

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANA SANGAMA, BELAGAVI- 590 018



“AICTE Activity Point Programme”

Submitted in Partial Fulfilment for the Award of Degree of
Bachelor of Engineering
in
Information Science and Engineering

Submitted by

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1HK22IS117

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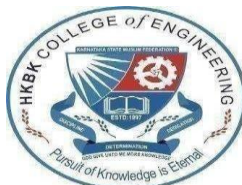
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AICTE Activity Points Programme CERTIFICATE

This is to certify that the **AICTE Activity Points Programme** has been successfully carried out by **Tamanna Tanwar H** bearing **USN – 1HK22IS117**, bonafide student of **HKBK College of Engineering** in partial fulfilment of the requirements for the award of degree of **Information Science and Engineering** of **Visvesvaraya Technological University, Belagavi** during **2022 – 2026**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report. The AICTE Activity Points Programme report has been approved as it satisfies the academic requirements in respect of AICTE Activity Points Programme for the said degree.

.....
Signature of Faculty In-charge
(Prof. Syeda Ayesha Amreen /
Prof. Sneha K)

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(Dr. A. Syed Mustafa)

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**Signature of NSS Officer / AICTE
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(Dr. Arshad Pasha / Prof. Sneha K)

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(Dr. Mohammed Riyaz Ahmed)

CERTIFICATE PHOTOCOPY

BBMP CERTIFICATE



BRUHAT BENGALURU MAHANAGARA PALIKE

Date: 10/01/26

This is to certify that **Tamanna Tanwar** bearing USN: **1HK22IS117** studying in HKBK College of Engineering has satisfactorily completed AICTE activity on topic **FACILITATING 100% DIGITALIZED MONEY TRANSACTIONS** in locality Which comes under **BRUHAT BENGALURU MAHANAGARA PALIKE (BBMP)**

Location:

4th cross Vayalikaval HBCS layout, Nagawara, Bangalore.

Pin code: 560045.



BBMP SEAL AND SIGNATURE

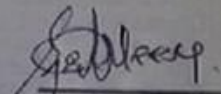
CERTIFICATE OF PARTICIPATION

This is to certify that Mr./Ms.
Tamanna Tanwar (1HK22IS117), *the student*
of

HKBK COLLEGE OF ENGINEERING
have conducted an awareness activity on
**Developing Sustainable Water
Management system**
at Ramanagaram
on 15 /11 /2025
as part of the AICTE Student Activity
Programme.

Syed Akhleem
City Training Council for
Villio No. 52, Yakkondan Mahala,
Ramanagaram Town - 582159
Mob: 9828860032

OFFICIAL SEAL



SIGNATURE



ಬೃಹತ್ ಬೆಂಗಳೂರು

ಮಹಾನಗರ ಪಾಲಿಕೆ

Bruhat Bengaluru Mahanagara Palike

AICTE Activity Programme

This activity is to certify that the below students of HKBK College of Engineering studying in 7th semester have successfully completed the outreach programme on "contribution to national level initiative of government of India (Swachh Bharat)" at Nagawara, Bengaluru, Karnataka. This programme was effective in providing the education of cleanliness, maintenance of garbage, disposal of wet and dry waste, segregation of waste to produce manure and importance of maintaining a healthy environment.

R Afshan Anjum

Tamanna Tanwar

Yealena Barman

CHIEF EXECUTIVE OFFICER
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LORDS OF CLIMATE

BBMP Seal and Sign

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Energy Swaraj Foundation
Energy by Locals for Locals

CERTIFICATE

OF ENERGY LITERACY

THIS CERTIFICATE IS PRESENTED TO

Tamanna Tanwar

has successfully completed the course spanning over 12 modules. The training helps people in understanding energy, its generation, consumption, wastage, carbon footprint, impact on the environment, means to avoid and minimize energy usage, alternative energy solutions, ways to become carbon neutral, misconceptions on solar energy, and approach to adopt solar energy solutions. This training is a part of the "ENERGY LITERACY TRAINING" of the Energy Swaraj Foundation.

01/01/2026

Date

Funding Partners



Prof. Chetan Singh Solanki

The authenticity of this certificate can be validated at <https://www.es-gaf.org/verifyCertificate>

CERTIFICATE

OF COMPLETION

GOVT HIGHER PRIMARY SCHOOL

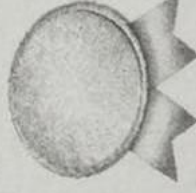
This Is To Certify

Tamanna Janwar [1HK22IS117]

have taught the importance resource utilization ,experience learning and Education
which was conducted for our students done at GOVT.HIGHER PRIMARY SCHOOL,GOVINDPUR
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DECLARATION

I, **Tamanna Tanwar H** bearing the **USN:1HK22IS117**, student of Bachelor of Engineering, **Information Science and Engineering, HKBK College of Engineering, Bengaluru**, hereby declare that the **AICTE Activity Points Programme** work has been carried out by me under the supervision and guidance of Department Staff Coordinators submitted by me as a partial fulfilment for the award of **Bachelor of Engineering** degree in **Information Science and Engineering** from **Visvesvaraya Technological University, Belagavi** during **2022 – 2026**.

I hereby declared that the AICTE activity point work has been carried out at following partner organizations:

1. Kadhamanthale male forest, Bengaluru
2. Sreemanhalli, Bengaluru
3. Dobbaspeth village, swacha sankeerna, Bangalore rural
4. Nelamangla Taluka, Bengaluru Rural District
5. Arakere Panchayathi, Bangalore

[Tamanna Tanwar H 1HK22IS117](#)

ACKNOWLEDGEMENT

I extend my heartfelt gratitude to the **Mr. C.M. Ibrahim**, Chairman, **Mr. C.M. Faiz Mohammed** Director of HKBK College of Engineering along with **Dr. Mohammed Riyaz Ahmed**, Principal of HKBK College of Engineering.

I am deeply indebted to **Dr. A. Syed Mustafa, Professor & Head**, Information Science and Engineering for the ineffable encouragement he provided in successful completion of the AICTE Activities and Programme.

Prof. Sneha K, College AICTE Activity Points Coordinator at HKBK College of Engineering, and **Prof. Syeda Ayesha Amreen**, Assistant Professor and Departmental Faculties in-charge at HKBK College of Engineering, for their unwavering support and provision of essential resources for the successful execution of this program.

I am sincerely thank our guide, **Prof.Sneha K**, Assistant Professor for their constant assistance, support, patience, endurance and constructive suggestions for the betterment of the Activity.

I am also deeply thankful to **all the NGOs and Organizations** for their invaluable support in offering meaningful volunteering opportunities and resources.

Lastly, I would like to express my sincere appreciation and respect to my parents, the teaching and non-teaching staff of the Department, and all my friends who have supported me, directly or indirectly, throughout my AICTE Activity Point Programme journey.

Tamanna Tanwar H - 1HK22IS117

Work Content

Sl. No.	Activity Heads	From date – To date	Number of Week	Total No of Hours	Performance Appraisal Points earned (To be Filled by Coordinators)	Remarks
1	Helping local schools to achieve good result & enhance their enrolment in Higher/ Technical/ Vocational Education					
2	Preparing an actionable business proposal for enhancing the village income					
3	Developing Sustainable Water Management system	01/03/2025 – 25/03/2025	3	80		
4	Tourism Promotion Innovative Approaches					
5	Promotion of Appropriate Technologies	13/07/2025 – 25/1/2025	2	80		
6	Reduction in Energy Consumption					
7	To Skill rural population					
8	Facilitating 100% Digitized money transactions					
9	Setting of the information imparting club for women leading to contribution in social and economic issues					
10	Developing and managing efficient garbage disposal system	03/08/2025 – 31/08/2025	2	80		
11	To assist the marketing of rural produce					
12	Food preservation/packaging					
13	Automation of local activities					
14	Spreading public awareness under rural outreach programmes	25/11/2025 – 20/12/2025	3	80		
15	Contribution to any national level initiative of Government of India, For e.g. Digital India/ Skill India/ Swachh Bharat Internish etc	01/01/2025 – 30/01/2025	4	80		
16	Creating an awareness regarding rain water harvesting in urban and rural area					
TOTAL ACTIVITY POINTS EARNED						

ABSTRACT

The AICTE Activity initiative aimed to address key societal challenges through five impactful projects focusing on education, sustainability, digital transformation, water conservation, and tourism promotion. The primary objectives included improving academic performance and vocational education enrolment, enhancing waste management practices, promoting cashless transactions, creating awareness about rainwater harvesting, and leveraging innovative strategies for tourism growth. Each activity was meticulously planned and executed through a combination of awareness campaigns, community engagement, digital literacy workshops, and hands-on implementation of sustainable practices. Efforts in schools led to improved student outcomes and increased vocational education participation. Waste management initiatives resulted in cleaner public spaces and higher community participation in recycling. The promotion of digital transactions enhanced financial literacy and reduced cash dependency, fostering a more secure and inclusive financial ecosystem. Rainwater harvesting awareness programs encouraged sustainable water management, mitigating urban and rural water scarcity. The tourism promotion initiative utilized digital content and community outreach to attract visitors and boost local economies.

The outcomes of these activities were significant, demonstrating tangible improvements in their respective focus areas. Academic achievements improved, digital transactions saw higher adoption, environmental cleanliness was enhanced, sustainable water conservation practices were encouraged, and tourism engagement increased. These initiatives collectively contributed to economic development, social empowerment, and environmental sustainability.

Moving forward, the activities hold great potential for expansion through policy integration, technological advancements, and continued community involvement. Strengthening collaborations with local authorities, businesses, and educational institutions can ensure the long-term success and scalability of these initiatives, further amplifying their positive impact on society.

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CHAPTER – 1

INTRODUCTION

Community-centered engineering practice converts classroom learning into measurable social value. This AICTE activity program was conceived and executed to address interlinked challenges in water security, livelihoods, digital inclusion, education access, and technology adoption. The initiative combined field drives, business- and skills-workshops, school outreach, technology scouting, and institutional visits so that practical solutions could be piloted and documented for scale. Importantly, every activity in this program was executed as a 90-hour work package and carried out entirely by a dedicated team of 8 students (no faculty accompanied the team during field implementation). The student-led model emphasized ownership, peer learning, rapid iteration, and sustained documentation all aligned with AICTE’s learning-through-service objectives.

1.1 Vision, Rationale and Objectives

Engineering education produces problem-solvers; when those problem-solvers engage deliberately with communities, the outcome is twofold: measurable community benefit and deeper student learning. The program’s core vision was to create reproducible, low-cost interventions that improve local welfare while offering students authentic project experience from needs assessment to monitoring and reporting.

Primary rationale:

- Rural communities frequently face seasonal water shortages, limited market access for micro-enterprises, low awareness about higher/technical education, and constrained exposure to appropriate technologies.

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- Small, well-designed interventions when combined with local governance buy-in and follow-up mentorship can create durable change.
- Student teams, when trained in participatory methods and given clear 90-hour work packages, can implement, document and hand over pilots that local actors can sustain.

Key objectives:

1. Improve local water resilience through rain water harvesting awareness and demonstration.
2. Increase rural incomes by co-creating actionable micro-business models and delivering digital-payments and e-listing training.
3. Raise student awareness of higher, technical and vocational pathways and support early application steps.
4. Scout and shortlist appropriate technologies for rural pilots by participating in national technology forums.
5. Expose students to research and industrial best practices via institutional visits to inform project design.
6. Produce an AICTE-ready, evidence-backed activity report.

1.2 Team Composition, Workload and Student-led Modality

All field activities were executed by a cohesive team of 8 student volunteers during visits, workshops, or demonstrations; faculty involvement was limited to prior coordination, approvals and post-activity technical review. This intentional design demonstrated that well-trained students can plan and implement community projects under appropriate institutional governance and local stakeholder support.

Workload model (exact, per-activity):

- Each activity was scoped as a 90-hour work package. The 90 hours covered planning, stakeholder coordination, field delivery, post-activity documentation, and initial

monitoring. Typical distribution for a single activity:

- Planning & stakeholder coordination: 12–15 hours
- Field delivery (sessions, demonstrations, installations, visits): 28–36 hours
- Documentation & evidence collection (attendance sheets, photos, minutes): 10–12 hours
- Post-activity follow-ups & synthesis (M&E templates, action notes): 20–25 hours
- Buffer for logistics / unforeseen tasks: 5–10 hours

Because each activity carried the same 90-hour commitment, the programme preserved parity across different intervention types (technical demo, business coaching, school outreach, summit scouting, or factory visit). Students organised themselves into sub-teams (planning, delivery, documentation, M&E) and rotated roles across activities to ensure broad exposure and equitable workload.

1.3 Swachh Bharat

The central objective of this activity was to promote cleanliness, hygiene awareness, and civic responsibility within the community. The initiative aimed to encourage behavioral change toward maintaining clean surroundings and reducing environmental pollution through active public participation in national cleanliness efforts.

- **Cleanliness Awareness Drives:** Participated in community campaigns promoting hygienic public spaces and responsible civic behavior.
- **Hygiene Education:** Educated residents about the importance of sanitation and personal hygiene in preventing diseases.
- **Community Engagement:** Encouraged active involvement in keeping streets and public areas clean.

Alignment with SDGs:

- **SDG 3 – Good Health and Well-being:** Promoted hygienic practices to reduce health risks.

- **SDG 11 – Sustainable Cities and Communities:** Supported cleaner and safer living environments.
- **SDG 12 – Responsible Consumption and Production:** Encouraged proper waste handling practices.

1.4 Technology & Institutional Exposure Bangalore TechSumit, CDAC Electronic City

Parallel to field work, the student team took part in technology scouting and institutional visits to broaden the program's technical options and inform pilot design. Participation at the national technology summit enabled rapid vendor engagement and produced a prioritized shortlist of affordable and field-appropriate technologies (soil-moisture sensors, low-power refrigeration monitors, solar DC micro-solutions). The visit to the research institution provided insight into data-driven project design and potential collaboration pathways for student projects.

1.5 Expected Outcomes and Alignment with SDGs

By design, each 90-hour activity produced deliverables that could be validated and cited in the AICTE report: attendance sheets, signed panchayat pledges, action notes, one-page business proposals, technology briefs, photographic evidence with captions, and initial M&E data. Collectively, the programme maps clearly to these Sustainable Development Goals:

- **SDG 3 (Good Health and Well-being):** Supported disease prevention and child health.
- **SDG 6 (Clean Water & Sanitation):** RWH demonstrations and panchayat integration for recharge practices.
- **SDG 8 (Decent Work & Economic Growth):** business proposals and digital-payments adoption for local entrepreneurs.

- **SDG 9 (Industry, Innovation & Infrastructure):** technology scouting, CDAC alignment and pilot design.
- **SDG 11 (Sustainable Cities and Communities):** Supported sustainable technological integration.
- **SDG 12 (Responsible Consumption and Production):** Promoted sustainable waste practices.
- **SDG 13 (Climate Action):** Supported environmental sustainability through conservation.
- **SDG 17 (Partnerships for the Goals):** Contributed through collaborative community efforts.

CHAPTER – 2

DETAILS OF ACTIVITIES

2.1 Contribution to any National Level Initiatives of Government of India

ACTIVITY HEAD: Swachh Bharat

Objective: The initiative sought to instill a strong sense of civic responsibility among residents and motivate them to actively contribute to maintaining sanitary conditions in both private and public spaces.



Fig 2.1: Team Spreading Awareness about Swachh Bharat.

Background & Rationale

Improper waste disposal, open dumping, and lack of community awareness contribute significantly to environmental degradation, soil contamination, and public health risks in peri-urban and semi-rural areas. Despite government efforts under the Swachh Bharat Mission, sustained cleanliness requires active community participation,

behavioral change, and practical demonstration of waste management practices.

Educational institutions play a critical role in promoting civic responsibility among students and communities. Awareness campaigns alone often fail to produce long-term behavioral change unless supported by visible action and community engagement. Therefore, this activity combined hands-on cleanliness drives, waste collection, awareness sessions, and community sensitization to promote hygiene, environmental sustainability, and responsible waste disposal practices.

Beneficiaries

Primary: Local residents (~60–100 individuals) living near the identified public space

Secondary: Adjacent residential clusters benefiting from improved environmental hygiene.

Detailed Implementation (90-hour workplan exact allocation)

- **Planning & coordination – 12 hours**

Student core team identified the cleanup site, coordinated with local authorities, and obtained necessary permissions. Roles were assigned, safety measures discussed, and activity schedule finalized.

- **Site survey & baseline assessment – 8 hours**

Students visited the location to assess waste types (plastic, paper, organic waste), volume estimation, and environmental impact. GPS-tagged photographs and site documentation were collected.

- **Awareness preparation & material development – 10 hours**

Students prepared awareness posters, placards, and educational material explaining waste segregation, recycling, and environmental protection in local language.

- **Community awareness sessions – 12 hours**

Four awareness sessions were conducted involving residents, explaining waste

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management, importance of cleanliness, and encouraging community participation in maintaining hygiene.

- **Cleanliness drive execution – 20 hours**

Students conducted hands-on cleaning activities including:

- Collection of plastic, paper, and solid waste
- Segregation of recyclable and non-recyclable waste
- Safe disposal and handover to municipal waste services
- Clearing dry leaves and debris

- **Waste segregation demonstration – 6 hours**

Students demonstrated proper waste segregation methods and explained biodegradable vs non-biodegradable waste management.

- **Safety, hygiene training & handover – 6 hours**

Participants were trained on safe waste handling, personal hygiene, and preventive measures. Local volunteers were identified for future monitoring.

- **Documentation & reporting – 8 hours**

Attendance sheets, photographs, GPS-tagged evidence, activity reports, and feedback were collected and compiled.

- **Follow-up planning & monitoring setup – 8 hours**

Students identified local volunteers and established a simple reporting system to monitor cleanliness and ensure sustainability.

Total: 90 hours (8 students organized into rotating sub-teams)

Materials & Technology Used

- Gloves, masks, garbage bags, and cleaning tools.
- Waste segregation bins.
- Awareness posters and printed leaflets.

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- GPS-enabled smartphone for documentation.
- Basic safety and sanitation equipment.
- Camera for activity recording and reporting.

Team Roles & Timeline (8 students, no faculty)

- **Core Coordination Team (2 students):**
Planning, permissions, and overall coordination
- **Technical & Cleanup Team (2 students):**
Waste collection, segregation, and disposal
- **Community Awareness Team (2 students):**
Conducting awareness sessions and educating residents
- **Documentation Team (2 students):**
Photography, attendance records, GPS tagging
- **Monitoring & Follow-up Team (2 students):**
Volunteer coordination and follow-up tracking

Timeline: Executed over a 2-week period including preparation, field execution, and follow-up.

2.2 Developing Sustainable Water Management System

ACTIVITY HEAD: Water Management

Objective: The primary objective of the Sustainable Water Management activity was to create awareness about the importance of conserving water and adopting efficient water usage practices.



Fig 2.2: Water Management Activity

Background & Rationale

Water scarcity and declining groundwater levels are major environmental and socio-economic challenges in many parts of India, especially in peri-urban and semi-rural regions. Rapid urbanization, excessive groundwater extraction, inadequate rainwater harvesting, and poor water management practices have contributed to seasonal water shortages and increased dependency on tanker water supply.

Beneficiaries

Primary: Community members dependent on groundwater sources

Secondary: Local governing bodies responsible for water resource management

Detailed Implementation (90-hour workplan)

- **Planning & stakeholder coordination – 12 hours**

Student core team identified suitable locations, coordinated with local authorities and community representatives, and prepared implementation plans and permission documentation.

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- **Site survey & baseline assessment – 10 hours**
Students conducted field surveys to assess water availability, existing infrastructure, groundwater conditions, and potential recharge locations. GPS-tagged photos and water usage observations were recorded.
- **System design & technical planning – 12 hours**
Technical team designed a sustainable water management system including rainwater harvesting structures, filtration units, and recharge pits. Materials and construction requirements were finalized.
- **Awareness and education sessions – 10 hours**
Students conducted awareness programs explaining water conservation methods, sustainable practices, and benefits of rainwater harvesting.
- **Material procurement & logistics – 8 hours**
Students procured necessary materials such as PVC pipes, filtration media (sand, gravel), storage containers, and construction tools from local vendors.
- **Installation & implementation – 18 hours**
Students assisted in installation of sustainable water management components including:
 - Rainwater collection and filtration systems
 - Groundwater recharge pits
 - Basic water monitoring setup
 - Demonstration models
- **System testing & demonstration – 6 hours**
Students tested system functionality and demonstrated water flow, filtration, and recharge process to community members.
- **Documentation & reporting – 8 hours**
Students documented activities through photographs, attendance sheets, GPS tagging, and technical records.
- **Monitoring setup & follow-up planning – 6 hours**

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Students established simple monitoring procedures and trained community volunteers for maintenance and observation.

Total: 90 hours

Materials & Technology Used

- PVC pipes and connectors
- Rainw water collection components (gutters, filters)
- Sand, gravel, and filtration media
- Recharge pit construction materials
- Water storage containers
- Monitoring sheets and documentation tools
- Smartphone for GPS-tagged documentation
- Basic construction and safety tools

Team Roles & Timeline (25 Students, No Faculty)

- **Core Coordination Team (4 students):**
Planning, permissions, and stakeholder communication
- **Technical Team (7 students):**
System design, measurements, and implementation
- **Community Awareness Team (5 students):**
Conducted awareness programs and education sessions
- **Documentation Team (4 students):**
Activity recording, reporting, and GPS documentation
- **Monitoring & Evaluation Team (3 students):**
Follow-up planning and monitoring system setup
- **Logistics Team (2 students):**
Material procurement and transport coordination

Timeline:

Activity completed over a 2-week implementation period including planning, installation, and follow-up.

2.3 Promotion of Appropriate Technologies

ACTIVITY HEAD: Promotion Of Appropriate Technologies

Objective: The primary objective of this activity was to promote awareness and understanding of appropriate technologies that can support rural development and sustainable growth.

Fig 2.3: Team Promoting Plastic Ban.



Background & Rationale

Many communities, particularly in semi-urban and rural areas, face challenges due to lack of access to affordable, simple, and sustainable technological solutions. High-cost and complex technologies are often unsuitable because they require specialized maintenance, electricity, or expensive infrastructure.

Beneficiaries

Primary: Students involved in learning sustainable technology applications.

Secondary: Local community leaders and institutions.

Detailed Implementation (90-hour workplan)

- **Planning & coordination – 14 hours**

The student team identified community needs, selected appropriate technologies, and coordinated with local representatives for activity planning and permissions.

- **Community survey & needs assessment – 12 hours**

Students conducted field visits to identify key problems such as water wastage, improper waste handling, and lack of energy-efficient solutions. Data was recorded through observation and interviews.

- **Awareness and training sessions – 12 hours**

Students conducted awareness sessions explaining appropriate technology concepts, benefits, and practical applications using demonstrations and visual materials.

- **Material procurement & preparation – 8 hours**

Students procured locally available, low-cost materials required for demonstration models and technology implementation.

- **Demonstration & implementation – 14 hours**

Students installed and demonstrated appropriate technology solutions and explained their working, benefits, and maintenance procedures.

- **Documentation & reporting – 8 hours**

Students documented activities through photographs, attendance records, GPS tagging, and prepared activity reports.

- **Monitoring & follow-up planning – 8 hours**

Students trained local volunteers on usage and maintenance and developed a follow-up monitoring plan.

Total: 90 hours (8 students working collaboratively with shared responsibilities)

Materials & Technology Used

- Laptops, projector, role-model videos stored offline.

Team Roles & Timeline (25 students)

- **Team Leader (1 student):**

Overall coordination, planning, and communication

- **Technical Team (3 students):**

Technology design, model preparation, and demonstrations

- **Community Awareness Team (2 students):**

Conducted awareness sessions and community interaction

- **Documentation & Monitoring Team (2 students):**

Photography, reporting, and follow-up planning

Timeline: Activity completed over a 2-week period including planning, implementation, and monitoring setup.

2.4 Developing and Managing Efficient Garbage Disposal System

ACTIVITY HEAD: Waste Segregation Drive

Objective: The primary objective of the Waste Segregation Drive was to promote structured waste management practices within the community by encouraging segregation of waste into appropriate categories at the source level.



Fig2.4: Compost generation.

Background & Rationale

Improper garbage disposal and lack of efficient waste management systems contribute significantly to environmental pollution, public health risks, and unsanitary living conditions. In many rural and semi-urban areas, improper waste segregation and disposal lead to soil contamination, foul odors, and increased risk of disease transmission.

Beneficiaries

Primary: Panchayat waste management workers

Secondary: Nearby residential areas benefiting from improved waste management.

Detailed Implementation (90-hour workplan)

- **Planning & coordination – 14 hours**

The student team coordinated with local panchayat authorities and waste management personnel to identify waste disposal facilities and plan the activity.

- **Waste management system study & analysis – 14 hours**

Students studied waste collection, segregation, storage, and disposal processes. They identified areas for improvement and prepared recommendations.

- **Awareness and training sessions – 12 hours**

Students conducted awareness programs for community members explaining waste segregation, recycling, and importance of proper garbage disposal.

- **Demonstration of waste segregation – 12 hours**

Students demonstrated proper segregation of waste into biodegradable, non-biodegradable, and recyclable categories using color-coded bins.

- **Documentation & reporting – 10 hours**

Students documented activities using photographs, attendance sheets, GPS-tagged images, and prepared reports.

- **Monitoring and maintenance planning – 8 hours**

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Students assisted in developing a monitoring plan and encouraged community participation in maintaining proper waste disposal practices.

- **Follow-up and evaluation – 8 hours**

Students conducted follow-up discussions with local authorities and community members to evaluate effectiveness and sustainability.

Total: 90 hours (8 students working collaboratively in structured roles).

Materials & Technology Used

- Color-coded waste segregation bins

Team Roles & Timeline (25 students)

- **Team Leader (1 student):**

Activity coordination and authority communication

- **Technical & Waste Management Team (3 students):**

Waste system study, segregation demonstration, and implementation

- **Documentation & Monitoring Team (2 students):**

Activity recording, reporting, and monitoring setup.

Timeline: Activity conducted over a 2-week period including planning, implementation and follow ups.

2.5 Spreading Public Awareness Under Rural Outreach Programmes

ACTIVITY HEAD: National Pulse Polio Program

Objective: The primary objective of participating in the National Pulse Polio Programme was to promote awareness regarding immunization and encourage families to ensure timely vaccination for children. Public awareness programmes play a critical role in educating communities, encouraging behavioral change, and empowering individuals to adopt responsible practices. Student-led rural outreach programmes help bridge the gap between knowledge and implementation by providing direct interaction, demonstrations, and educational support.

Fig 2.6: Promoting Pulse Polio Camp



Background & Rationale

Rural communities often face challenges related to lack of awareness about sanitation, environmental protection, health, water conservation, and sustainable living practices. Limited access to information and educational resources can prevent rural populations from adopting practices that improve their quality of life and environmental sustainability.

Beneficiaries

Primary Beneficiaries: Students gaining practical exposure to rural outreach activities.

Secondary Beneficiaries: Nearby rural communities benefiting from awareness spread.

Detailed Implementation (90-hour workplan)

- **Planning & coordination – 14 hours**

The student team coordinated with village authorities and local representatives to plan outreach activities, obtain permissions, and identify awareness topics relevant to the community.

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- **Community survey & needs assessment – 12 hours**

Students conducted surveys and interacted with local residents to identify awareness gaps related to sanitation, water conservation, waste management, and hygiene.

- **Awareness material preparation – 12 hours**

Students prepared educational materials such as posters, banners, pamphlets, and presentation content in local language for effective communication.

- **Awareness campaign & public interaction – 16 hours**

Students conducted awareness campaigns through direct interaction, demonstrations, and explanation of important practices such as proper waste disposal, hygiene, and water conservation.

- **Documentation & reporting – 10 hours**

Students documented activities through photographs, attendance records, and GPS-tagged images, and prepared detailed reports.

- **Monitoring & follow-up planning – 8 hours**

Students developed a follow-up plan and encouraged community members to continue adopting the recommended practices.

Total: 90 hours (8 students actively participating in structured roles)

Materials & Technology Used

- Awareness banners and poster.

Team Roles & Timeline (25 students)

- **Team Leader (1 student):**

Activity planning, coordination, and communication with authorities

- **Awareness & Outreach Team (3 students):**

Conducted awareness sessions and community interaction

- **Technical & Demonstration Team (2 students):**

Demonstrated sustainable practices and provided guidance

- **Documentation & Monitoring Team (2 students):**

Recorded activities, collected evidence, and planned follow-up

- **Timeline:**

Activity completed over a 2-week period including planning, awareness campaigns, and follow-up planning.

CHAPTER – 3

OVERALL LEARNINGS OF THE ACTIVITIES

This chapter synthesizes cross-cutting lessons, insights and reflections drawn from the full program of student-led interventions. All activities were implemented as discrete 90-hour work packages, executed by a cohesive team of 8 students without faculty accompaniment during field delivery. The chapter condenses operational learning (what worked / didn't), technical and social insights, measurable educational outcomes, capacity-building for students, sustainability lessons, and clear recommendations for future replication and scale-up.

3.1 Summary of what was learned

- Student-led implementation at scale is feasible. A motivated cohort of 8 students planned, executed and documented multiple activities (community RWH, business skilling, school outreach, tech-scouting, institutional & industrial visits) and produced AICTE-grade evidence (attendance sheets, signed panchayat pledges, action notes, photos and transaction logs).
- Equal time-boxing (each activity = 90 hours) simplified planning, ensured parity of effort, and created a repeatable “unit” of community engagement that is easy to budget, evaluate and replicate.
- Community buy-in (panchayat leadership, local champions) was decisive: permissions, participant turnout and post-handover maintenance depended heavily on local ownership.
- Low-capex, modular solutions produce quicker adoption than high-tech, high-capex alternatives. Vendors with prior rural deployments and local service networks were prioritized during tech-scouting.

- Evidence and storytelling matter: small measurable wins (first UPI sale, pledge signed, demo installation finished) created momentum and unlocked follow-on interest and funding conversations.

3.2 Operational & Project-management learnings

- **Advance coordination reduces friction:** Pre-work that included permission letters, a short local-language leaflet and a clearly defined itinerary significantly increased turnout and reduced on-site confusion.
- **Role clarity and rotation works:** Students organized into functional pods (planning, delivery, documentation, M&E). Rotating roles across activities helped distribute experiential learning and avoided burnout.
- **90-hour scoping is a powerful unit:** The predefined 90-hour package forced realistic scoping: students learned to prioritize high-impact, low-effort deliverables and to document evidence within time constraints.
- **Documentation must be continuous, not deferred:** Real-time photo capture, attendance signature, and immediate uploading to a shared folder avoided the common pitfall of “lost” evidence at the end of fieldwork.
- **Logistics and procurement require a small dedicated team:** A 3-student logistics sub-team that handled local vendor negotiation, transport and petty cash avoided last-minute delays and kept costs lower.

3.3 Community engagement & social learnings

- **Local leadership is the accelerator:** Where panchayat members actively endorsed the event, attendance, trust and post-activity commitments were higher. The signed panchayat pledge for RWH is an example of a low-cost governance enabler.
- **Trust-building through demonstration beats theory:** Hands-on installations (RWH demo pits) and live role plays (digital payment transactions) converted sceptics faster than classroom presentations.

- **Adapt messaging to local context:** Visuals, vernacular-language leaflets and simple physical models were far more effective than technical slides. For school sessions, role-model stories and local examples resonated best.
- **Economic incentives matter:** For entrepreneurs, immediate cost-savings or plausible revenue increase (even if small) was the principal motivator for adoption. Business proposals that showed quick payback were prioritized.

3.4 Technical & tool-level learnings

- **Prefer modular & maintainable tech:** Shortlisted technologies from the summit and vendor demos were favoured when they: (a) had simple maintenance steps, (b) could be serviced regionally, (c) were modular (so upgrades were optional), and (d) allowed staged investment.
- **Design for offline-first behaviour:** Several local contexts have intermittent connectivity. Tech solutions that cache data and operate offline with periodic sync (SMS/GSM fallbacks) were rated higher.
- **Cold-chain requires systems thinking:** The industrial visit to the dairy processing facility reinforced that reducing spoilage is not a single tech fix but an aggregation of logistics, chilling, packaging and trust (timely payments and aggregation points).

3.5 Educational & capacity-building learnings

- **Project ownership accelerates learning:** Students reported rapid gains in soft skills (community facilitation, negotiation, empathy), hard skills (basic civil construction supervision, digital payments, product-photography) and project management.
- **Peer-led mentorship scales:** The mentor chain established for school students (8 student mentors for ~20 interested pupils) proved manageable and scalable with clear weekly check-ins.
- **Field exposure informs final-year projects:** CDAC visit and technology scouting seeded multiple idea briefs suitable for final-year or research projects, linking academic outcomes to community needs.

3.6 Monitoring & evaluation learnings

- **Simple indicators are sustainable:** Attendance, number of UPI activations, signed panchayat documents, first paid transactions, and photographic evidence were easy to collect and meaningful.
- **Triangulation improves credibility:** Combining self-reported revenue changes with anonymized transaction screenshots and random spot-checks gave more credible impact reporting.
- **Longitudinal follow-up is necessary for claims of impact:** Short-duration interventions produce immediate outputs; however, medium-term impacts (income increases, water-table change, school enrolment shifts) require 3–6 month follow-up with the same measurement instruments.

3.7 Challenges encountered

- **Limited household liquidity for upfront costs:**
Approach: bulk procurement, panchayat subsidy proposals, phased installation plans.
- **Intermittent internet & device access.**
Approach: offline materials, SMS fallbacks, shared village digital hub concept.
- **No central authority on-site created responsibility pressures.**
Approach (positive learning): students developed peer escalation protocols and used formal pre-approved letters / panchayat permissions to handle escalations.

3.8 Measured educational & social outcomes

- **Community engagement:** Four awareness sessions and two public RWH demonstrations; signed panchayat pledge for RWH inclusion in building approvals.
- **Digital inclusion & livelihoods:** 25 entrepreneurs trained; 12 vendors enabled for UPI; three one-page business proposals produced; early paid digital orders recorded for two vendors.
- **Education:** ~60 students reached in the government school; 20 students placed into a mentor follow-up chain with weekly check-ins.
- **Technology & institutional learnings:** Three shortlisted technologies with vendor commitments for demo units; two student project briefs linked to CDAC capabilities and potential supervision.
- **Industrial insights:** Action note for dairy aggregation and cold-chain drafted and shared with panchayat and two village milk groups.

3.9 Sustainability, institutionalisation & scale-up learnings

1. **Government linkage is essential:** Embedding small changes (RWH checks, aggregation points) into panchayat processes is more sustainable than relying solely on external grants.
2. **Shared infrastructure models reduce unit costs:** Community chilling points, shared packaging kiosks, and a village digital hub (one phone, one trained operator) lowered barriers to entry.
3. **Revenue reflows can sustain operations:** For example, a small fee on aggregation services can finance chiller maintenance; training local youth as paid operators can create micro-employment.
4. **Replication requires a compact “Replication Kit.”:** A kit includes material lists, vendor contacts, a 90-hour workplan, monitoring templates and a governance letter template for panchayats.

CHAPTER – 4

DOCUMENTATION OF ACTIVITIES, EVIDENCE & ANNEXURES

2.1 ACTIVITY 1: Contribution to any National Level Initiatives of Government of India

Location: Kadhamanthale male forest, Bengaluru



Fig 4.1: Swachh Bharat Awareness carried out by team.



Fig 4.2: Happy face of the team after complete activity for good cause.

AICTE Activity Point Programme

During this 90-hour student-led activity, our team of 8 students organized a cleanliness and waste management awareness drive under the Swachh Bharat Mission in collaboration with the local village panchayat at Bengaluru. The team conducted community awareness sessions to educate residents about the importance of proper waste segregation, hygienic surroundings, and responsible garbage disposal practices for improving public health and environmental sustainability. Practical demonstrations were carried out using color-coded bins to explain the segregation of biodegradable and non-biodegradable waste, along with guidance on safe disposal and recycling methods. Students actively participated in cleanliness drives at public areas and waste collection points, supervised the proper use of waste bins, and documented the existing waste management infrastructure. Household-level interactions were conducted to encourage residents to adopt daily cleanliness practices and proper waste segregation at source. The activity concluded with support and commitment from the panchayat to promote regular sanitation practices and strengthen waste management systems, ensuring long-term sustainability and alignment with the objectives of the Swachh Bharat initiative.

ACTIVITY 2: Developing Sustainable Water Management System

Location: Sreemanhalli, Bengaluru

Fig 4.3: Water Management System.



During this 90-hour student-led activity, our team of 8 students organized an awareness drive on Rainwater Harvesting (RWH) in collaboration with the village panchayat. We conducted community sessions explaining the importance of groundwater recharge, water conservation, and sustainable water usage practices. Practical demonstrations were carried out, including the installation of rooftop gutter systems and recharge pits at selected public locations. Students supervised material procurement, assisted in measurements, and documented the complete installation process. Household-level guidance was also provided to encourage adoption of low-cost RWH models. The activity concluded with a panchayat-level commitment to promote RWH in future construction projects, thereby ensuring long-term sustainability of the initiative.

ACTIVITY 3: Promotion Of Appropriate Technologies

Location: Dobbaspeta village, swacha sankeerna, Bengaluru rural



Fig 4.6: Team gathered for promoting technologies.

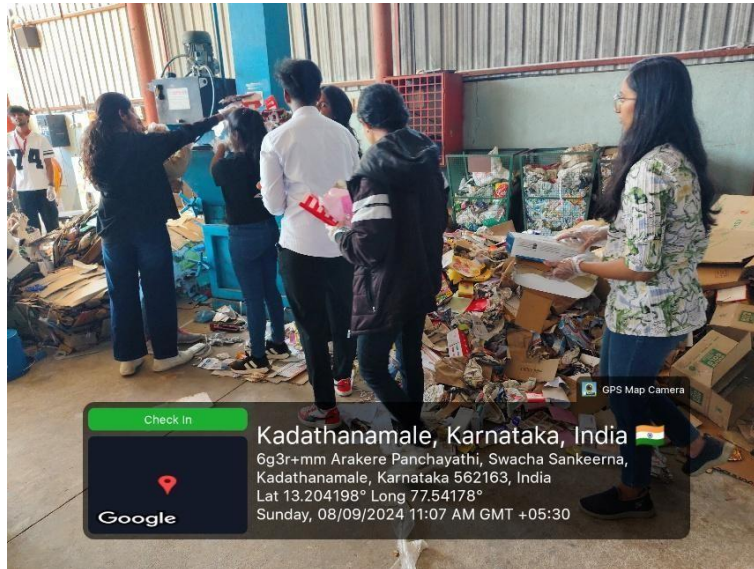


Fig 4.8: Carrying out activity.

During this 90-hour student-led activity, our team of 8 students organized an awareness and demonstration programme on promoting appropriate technology in collaboration with the local village panchayat at Bengaluru under the guidance framework of All India Council for Technical Education. The activity focused on introducing low-cost, sustainable, and locally adaptable technological solutions to address everyday community challenges related to water management, waste handling, and resource conservation. Students conducted interactive sessions explaining the importance of appropriate technology, highlighting its benefits such

ACTIVITY 4: Developing and Managing Efficient Garbage Disposal System

Location: Nelamangla Taluka, Bengaluru Rural District



Fig 4.9: Team gathered for Waste Segregation.



Fig 4.10: Helping In Waste Segregation.



Fig 4.11: Food Waste Segregation.

During this 90-hour student-led activity, our team of 8 students organized a Waste Segregation awareness and implementation drive in collaboration with the local village panchayat at Bengaluru under the Swachh Bharat Mission. The objective of the activity was to educate the community about the importance of segregating waste at the source to promote proper disposal, recycling, and environmental sustainability.

The team conducted awareness sessions explaining the different categories of waste, including biodegradable, non-biodegradable, and recyclable waste, and highlighted the environmental and health benefits of proper segregation. Practical demonstrations were carried out using color-coded bins to show the correct method of separating household waste. Students also visited public waste collection points to observe existing practices and assisted in improving segregation methods where necessary.

ACTIVITY 6: Spreading Public Awareness Under Rural Outreach Programmes

Location: Bangalore



Fig 4.15: Pulse Polio Awareness



Fig 4.16: Team gather

AICTE Activity Point Programme

In this 90-hour student-led rural outreach activity, our team of 8 students organized a public awareness campaign in collaboration with the local village panchayat at Bengaluru under the initiative of All India Council for Technical Education. The programme aimed to educate rural residents about essential topics such as sanitation, waste management, water conservation, hygiene, and sustainable environmental practices.

Students conducted interactive awareness sessions using posters, banners, and demonstrations to explain the importance of maintaining clean surroundings, proper waste disposal, and efficient use of natural resources. Household visits were also carried out to provide practical guidance and address community concerns related to sanitation and environmental sustainability. The team actively engaged with residents to encourage behavioral change and promote community participation in maintaining a healthy environment.

CHAPTER – 5

CONCLUSION

The AICTE Activity Programme undertaken and documented in this report represents a comprehensive, student-led initiative that successfully integrated community engagement, environmental sustainability, entrepreneurship development, educational awareness, technological exposure, and industrial learning. Each activity was executed as a structured 90-hour work package, carried out by a committed team of 8 students, without faculty accompaniment during field implementation. This format ensured equitable workload distribution, measurable outputs, and a replicable framework for future initiatives.

The programme demonstrated that structured student involvement can produce tangible community outcomes when guided by clear objectives, defined timelines, and systematic documentation. In Kammasandra village, the rainwater harvesting initiative increased awareness about groundwater recharge and led to the successful completion of public demonstration structures. The panchayat's commitment to integrating RWH checks into building approvals reflects institutional-level adoption and long-term sustainability.

The actionable business proposal and digital-skilling intervention strengthened rural economic participation. Local entrepreneurs were introduced to structured business planning, digital payments, and basic online market access tools. The adoption of UPI transactions and creation of WhatsApp Business catalogues signified a shift toward digital inclusion. These interventions align directly with national priorities such as Digital India and promote inclusive economic growth under SDG 8 (Decent Work and Economic Growth).

The institutional exposure visit to CDAC Electronic City deepened student understanding of high-performance computing, IoT architectures, and predictive analytics. This exposure informed the development of student project briefs aligned with research-

AICTE Activity Point Programme

oriented methodologies and enhanced appreciation for national technological capabilities.

A key strength of the programme was its evidence-driven approach. Each activity generated documented deliverables including attendance sheets, photographic records, signed acknowledgements, business proposals, technical briefs, and monitoring templates. This structured documentation not only strengthens the credibility of the report but also provides a replicable model for future AICTE activity implementations.

In summary, the AICTE Activity Programme successfully achieved its dual purpose: delivering meaningful community impact while enhancing experiential learning for students. The structured 90-hour per activity model, combined with strong documentation and measurable indicators, provides a scalable template for similar student- led initiatives across institutions. The collective outcomes reflect sustainable development, responsible innovation, and inclusive growth aligning closely with national development goals and AICTE's mission of fostering socially responsible engineers.

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